

TIC 59488906

PERIOD ALIAS

Eclipsing binary candidate dossier - Star Vetter

BLS EPHEMERIS

Period 13.937196 d (ALIASED, see below)
-> corrected period: 27.874392 d
Epoch (t0) 3236.2287 BTJD
Duration 0.2440 d
Depth 4.213 % (42,125 ppm)
SNR 3299.3

CATALOG CROSS-MATCH (novelty)

VSX	no match
ASAS-SN	no match
Gaia eclipsing-binary table	no match
Gaia variable classification	no match
TESS EB catalog (Prsa+2022)	no match
ExoFOP TOI list	no match

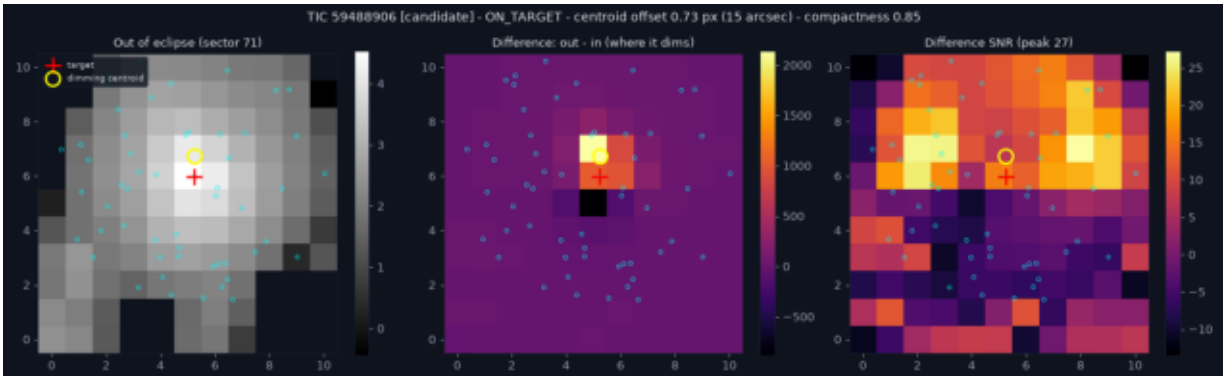
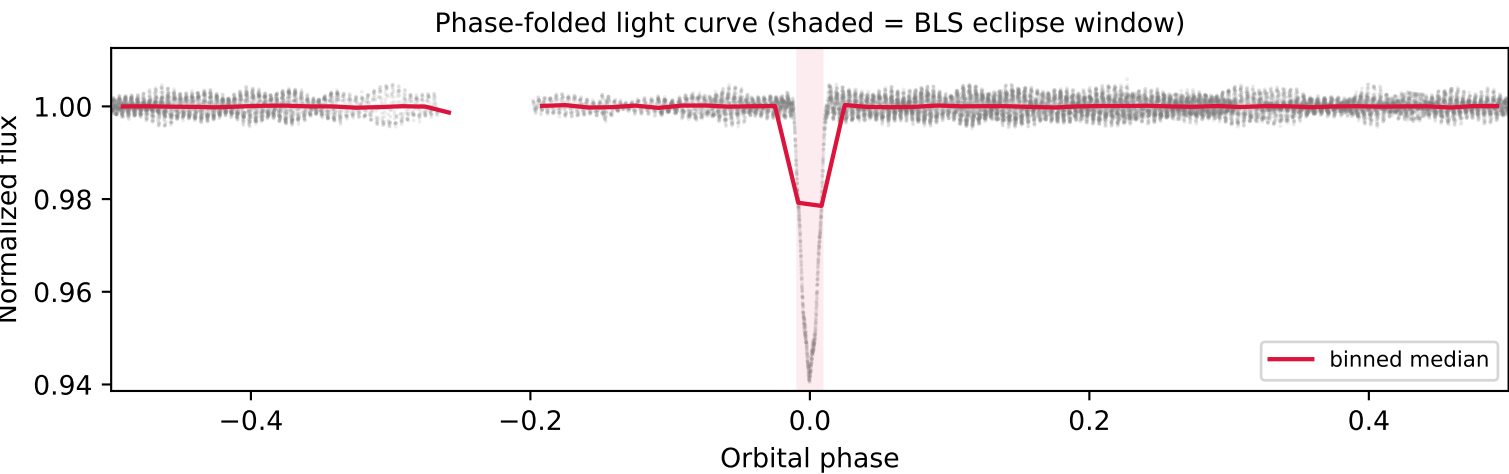
INDEPENDENT CHECKS (f3i gates)

Eclipses observed 4 (gate >= 3)
Robust depth ratio 1.08 (gate >= 0.50)
Robust depth sigma 398.3 (gate >= 7.0)
Odd/even z-score 7.62 (ALIAS FLAGGED)
Secondary eclipse not detected

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 71 light-curve dip in this sector: 248.1 sigma t0 refined by -0.0025 phase
Difference-image peak SNR 27.26 (gate >= 5.0) Compactness 0.85 (gate >= 0.50)
Dimming centroid offset 0.73 px (15 arcsec) (gate <= 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: none within the cutout



Caveat (PERIOD ALIAS): Odd/even eclipse-depth z-score is 7.62 (gate > 3.0): alternating eclipses differ significantly in depth, the signature of a true period double what BLS found. Pixel-check confirmation and eclipse timing are unaffected - report the corrected period, 27.874392 d, in any write-up.